

Literature Status Note

The Lipschitz-domain oscillation LIL from arXiv:1203.5290

Run `fa_banach_001` — lane 2

27 August 2026

**LITERATURE ALREADY ANSWERED — LIPSCHITZ-DOMAIN
BRANCH ONLY**

Source question

K. S. Eikrem, E. Malinnikova, and P. A. Mozolyako study harmonic functions on the upper half-space with growth controlled by a decreasing doubling weight v and prove a law of the iterated logarithm for

$$I_u(x, s) = \int_s^1 u(x, t) d\left(\frac{1}{v(t)}\right).$$

In Section 5, “Concluding remarks and open problems,” PDF page 19 of arXiv:1203.5290, they ask whether their Theorem 5 extends to Lipschitz domains [1]. In the same paragraph they state a separate, stronger local conjecture under the radial hypothesis $|u(x, t)| \leq Cv(t)$ for x in a positive-measure set.

Supporting answer

P. Mozolyako’s later paper explicitly identifies the source theorem and says: “Our main goal is to extend this result to the domains ... above the graphs of Lipschitz functions” [2, p. 2]. Its Theorem 1, on supporting PDF page 2, proves exactly that extension. If

$$\Omega_\phi = \{(x, y) : y \geq \phi(x)\},$$

where ϕ is Lipschitz, and u satisfies

$$|u(x, y)| \leq Kw(\text{dist}((x, y), \partial\Omega_\phi)),$$

then for

$$I(x, \delta) = \int_\delta^1 u(x, \phi(x) + y) d\left(\frac{1}{w(y)}\right)$$

one has

$$\limsup_{\delta \rightarrow 0} \frac{I(x, \delta)}{\sqrt{\log w(\delta) \log \log \log w(\delta)}} \leq C \|u\|_{w, \infty} \quad \text{for almost every } x \in \mathbb{R}^n.$$

The supporting authors knew they were answering the source question: the paper cites the source as [EMM], restates its upper-half-space theorem as Theorem A, and presents Theorem 1 as the requested extension. It also notes that the same conclusion holds for certain star-like Lipschitz domains.

Scope boundary

This identification does *not* close the source paragraph’s radial-only local conjecture. Theorem 2 of arXiv:1507.07455 proves a local result under the stronger non-tangential hypothesis

$$|u(x, y)| \leq Kw(y) \quad \text{whenever } |x - x_0| \leq My, \quad x_0 \in \Sigma.$$

Immediately afterward, on supporting PDF page 3, the author states that it is unknown whether this can be replaced by radial growth control. Therefore the durable classification is:

- Lipschitz-domain extension: fully answered in later literature.
- Radial-only local version on a positive-measure set: still open in the decisive supporting paper; no claim of resolution is made here.

Evidence and review recommendation

The identification used the arXiv source files and local PDFs for both papers, plus exact-phrases and keyword searches for the source question. The decisive cross-checks are source PDF page 19, supporting PDF page 2 (Theorem 1), and supporting PDF page 3 (the radial/non-tangential distinction).

Human review recommendation: classify only the Lipschitz-domain branch as an already-answered literature result; retain the radial local conjecture as unresolved unless a later exact source is found.

References

- [1] K. S. Eikrem, E. Malinnikova, and P. A. Mozolyako, *Wavelet decomposition of harmonic functions in growth spaces*, arXiv:1203.5290; published as *Wavelet characterization of growth spaces of harmonic functions*, J. Anal. Math. 122 (2014), 87–111.
- [2] P. Mozolyako, *Boundary oscillations of harmonic functions in Lipschitz domains*, arXiv:1507.07455 (2015).